Gourang Agrawal

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Summary

Experienced Java Developer with more than 4 years of experience in designing, developing, and deploying scalable, database-driven microservices-based applications using Java, Spring Boot, and RESTful Web Services in the BFSI, Insurance, and Pharmaceutical domains. Experienced in end-to-end feature ownership, processor integrations, and event-driven architectures. Skilled in performance optimization, design patterns, debugging complex distributed systems, and collaborating with cross-functional Agile teams.

Work Experience

Ciklum India Pvt. Ltd., ***Java Engineer II*** **Pune, India | August 2025 – Present**

Mastercard Prepaid Dec 2025 – Present

- Owned the **end-to-end design and development** of the Block/Suspend/Deactivate feature for payment instruments across **2 microservices** (channel + processor layer), serving Mastercard's global prepaid card programs.
- Created the **low-level design** — defined API contracts, database schema (**Oracle/JPA**), domain aggregates, state machines, and sequence flows — reviewed and approved by senior architects before implementation.
- Architected a **7-step processing pipeline** using **Chain of Responsibility** pattern (Initialize → LoadConfig → Enrich → CallProcessor → Translate → ErrorHandle → Terminate), making the flow extensible for future operations.
- Implemented **idempotency and exactly-once guarantees** using client-unique-key tracking, **Redis** duplicate detection, and aggregate state transitions (INITIATED → COMPLETED → FAILED) across distributed payment calls.
- Built **E6 processor integration** with **FreeMarker**-based payload transformation, database-driven multi-tenant routing (**Strategy pattern**), and standardized error translation from raw processor codes to API error contracts.
- Published **domain events** (blockInitiated, blockCompleted, blockFailed) for downstream audit, analytics, and event-driven consumers — following **CQRS** and event-sourcing principles.
- Authored **end-to-end acceptance tests** (**RestAssured** + embedded H2 + stub connector) simulating real processor responses for happy-path, 4xx, and 5xx scenarios — zero dependency on live E6 environments.

MongoDB - Insurance Domain Aug 2025 – Dec 2025

- Migrated **30+ PL/SQL stored procedures** into **Java Spring Boot-based services** using **MongoDB** and **Mongo Repositories**, improving policy and customer data processing efficiency.
- Applied **batch job optimizations** — chunk-based processing, parallel execution via **Executor Framework**, pagination, and transaction management — to boost premium calculation performance.
- Developed and optimized multiple **RESTful Web Services** for policy, claims, and reporting modules, enhancing scalability and frontend integration.
- Leveraged **MongoDB aggregation pipelines** for efficient transformation of large policy and customer datasets, reducing query latency.
- Improved performance by eliminating redundant loops, enabling **DTO mapping**, and using **BulkWriteCollector** for fast bulk updates.

Cognizant Technology Solutions, ***Software Engineer*** Pune, India | June 2022 – August 2025

Pharmaceutical Domain (AMS 2.0)

- Fixed the **Reprocess Request** issue, allowing users to reprocess individual methods instead of all at once, improving workflow efficiency by **40–50%** in German operations.
- Created and optimized several **RESTful APIs** for laboratory applications, improving data flow, performance, and system scalability.
- Resolved a **DAO-layer bug** to exclude deleted users from queries, ensuring real-time data consistency and accurate UI updates.
- Developed and maintained core modules for **data analysis and laboratory management** using **Java, Spring Boot, Maven, Oracle, and Angular**.
- Designed **Angular-based UI screens** for dynamic role assignments and integrated **REST API calls** for smooth backend communication.
- Implemented a **file upload feature** for MOL structures, improving user task efficiency by **30–40%**.
- Enhanced security by integrating **AES-256 encryption** with **Kerberos authentication** through keytab updates.

Technical Skills

Languages & Frameworks: Core Java, Java 8+, Spring Boot, Spring MVC, Spring security, JPA, Hibernate, JWT, Microservices, Design Patterns, SQL

Web & Database: REST APIs, Angular, React, MySQL, Oracle, MongoDB, Redis, H2

Tools & Platforms: IntelliJ IDEA, Git, Maven, Postman, Insomnia, Swagger/OpenAPI, Splunk, Sonar, JUnit 5, Mockito, Docker, Jenkins, CI/CD, Generative AI

Cloud & Methodologies: AWS, Azure, Kafka, Event-Driven Architecture, DDD, LLD, SOLID Principles, Agile/Scrum

Education

Gyan Ganga Institute of Technology & Sciences — Jabalpur, India
B.Tech in Mechanical Engineering; ***CGPA: 9.29/10.00*** — *Jul 2018 – May 2022*

Kendriya Vidyalaya — Jhagrakhand, India
All India Senior School Certificate Examination (PCM); ***GPA: 88.20/100*** — *Mar 2016 – Jul 2018*

Certificates

Microsoft Certified — *June 2023*
Azure Fundamentals (AZ-900)